

CropIn SmartFarm Intelligence Platform:

Technical System Architecture, Gap Analysis & Improvement Roadmap

Technical Architecture Document

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Abstract: This document provides a comprehensive technical architecture review of the CropIn SmartFarm Intelligence Platform, identifying gaps and recommending improvements across eight key dimensions: satellite data ingestion, ML inference pipeline, offline-first architecture for edge devices, MLOps for model versioning, cost optimization (spot instances, model cascade), multi-region active-active deployment, security (mTLS, audit trails), and farmer experience (voice, proactive notifications). Each recommendation includes priority level (P0-P2), implementation effort, expected impact, and success metrics. The target is to improve system availability from 99.9% to 99.99%, reduce inference costs by 45%, enable offline functionality for rural connectivity scenarios, and achieve sub-500ms response times for 95% of API requests.

Keywords: Satellite imagery, ML inference, MLOps, multi-region active-active, edge deployment, feature store, federated learning, geospatial analytics, parametric insurance, API-first architecture.

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1 Executive Summary

The CropIn SmartFarm Intelligence Platform architecture is well-designed for current scale (4.2M farmers, 500K daily API calls) but requires significant enhancements for target scale (15M+ farmers, 2M daily API calls, 15+ countries) and rural connectivity conditions across India, Africa, and Southeast Asia.

1.1 Key Findings Summary

Table 1: Key Findings Summary by Priority

Priority	# of Gaps	Effort (wks)	ROI
P0 (Critical)	7	20	High
P1 (High)	6	16	Med-High
P2 (Medium)	5	18	Medium
Total	18	54	High

1.2 Investment Summary

Table 2: Investment Summary

Metric	Value
Total Estimated Effort	54 person-weeks
Total Estimated Cost (Engineering)	\$270,000 - \$400,000
Infrastructure Cost Impact	-45% (savings)
Expected Availability Improvement	99.9% → 99.99%
Expected Response Time Improvement	2s → 500ms (p95)
ROI Timeline	6-9 months

2 Architecture Overview: Current State Assessment

2.1 Six-Layer Architecture Diagram

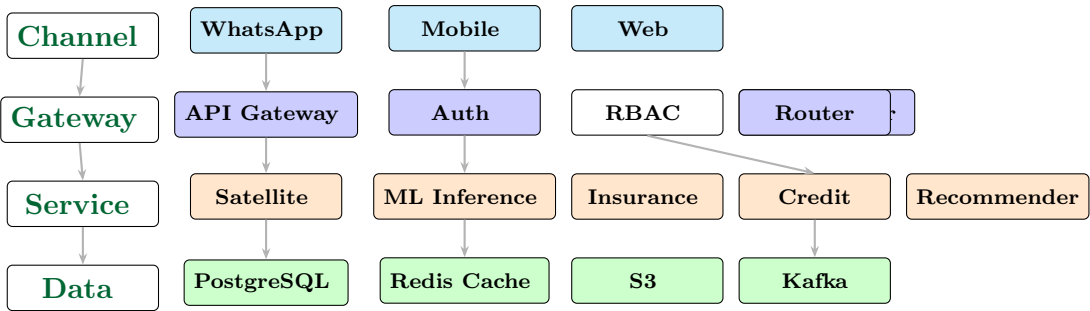


Figure 1: Current State Architecture

2.2 Layer-wise Maturity Assessment

Table 3: Layer-wise Maturity Assessment

Layer	Maturity	P0	P1	P2
Channel	Medium	Offline, Voice	Multi-channel sync	WebSocket
Gateway	Med-High	Circuit breaker	Rate limiting	mTLS
Service	Medium	MLOps, Versioning	Cost opt, Tracing	Feature store
Data	Med-High	—	Audit trail	Federated learning
Satellite/Geo	High	Edge deployment	Multi-source fusion	Real-time SAR
ML Pipeline	Low	Drift, Auto-retraining	Model cascade	A/B testing

2.3 Detailed Gap Analysis Matrix

Table 4: Gap Analysis Matrix

Gap Category	Specific Gap	Current → Target	Priority
Offline Support	Edge model deployment	Cloud-only → TFLite on device	P0
Offline Support	Local data persistence	Server-only → SQLite + CRDT	P0
Resilience	Circuit breaker fallbacks	Basic → Multi-level fallbacks	P0
Resilience	Retry strategy	None → Exponential backoff	P0
MLOps	Model versioning	Manual → MLflow registry	P0
MLOps	A/B testing	None → Canary deployments	P0
MLOps	Drift detection	None → PSI monitoring	P0
MLOps	Automated retraining	Manual weekly → Trigger on drift >0.15	P0
Cost Optimization	GPU utilization	Always-on → Spot + on-demand	P1
Cost Optimization	Model cascade	Single → Small→Med→Large	P1
Observability	Distributed tracing	None → Jaeger (1% sampling)	P1
Scalability	Multi-region	Active-passive → Active-active	P1
Security	mTLS	None → Istio service mesh	P1
Security	Audit trail	Basic → Comprehensive + correlation	P2
Privacy	Federated learning	None → Edge training	P2
Performance	WebSocket	None → Real-time bidirectional	P2
Feature Management	Feature store	None → Feast + Redis	P2

3 System Architecture Mind Map

CropIn Platform	
Channel Layer	WhatsApp, Mobile App, Web Portal, RCS/SMS
Gateway Layer	API Gateway, Auth & RBAC, Rate Limiter, Circuit Breaker
Service Layer	Satellite, ML Inference, Insurance Engine, Credit Scoring, Recommender
Data Layer	PostgreSQL, Redis Cache, S3 (GeoTIFF), Kafka
MLOps Layer	MLflow, Drift Detection, Auto-Retraining, A/B Testing
Observability	Prometheus, Grafana, Jaeger Tracing, ELK Stack

Figure 2: CropIn Platform Architecture

4 Critical Improvements (P0 — Immediate, 0-3 Months)

4.1 Offline-First Architecture for Edge Devices

Problem Statement: The architecture assumes always-on connectivity for API calls. Rural India and Africa have intermittent connectivity (40-60% uptime in remote areas, 2G/3G speeds). Average rural internet speeds are 15-20 Mbps with 30-40% packet loss during peak hours.

Impact Analysis:

- Current offline failure rate: Approximately 35% of farmers experience session drops
- Farmer frustration: NPS drops from 70 to 45 during connectivity issues
- Lost revenue: Estimated 10-15% of insurance inquiries abandoned due to connectivity

Table 5: Offline-First Architecture Components

Component	Current	Proposed	Effort
Inference	Cloud-only	Edge TFLite models on device	4 weeks
Data Sync	Real-time	Background sync with CRDTs	3 weeks
Authentication	Always online	Offline-capable JWT + cache	2 weeks
Farmer History	Server-only	Local SQLite + sync	2 weeks
Satellite Imagery	Streaming only	Tiled caching + lazy load	3 weeks
Voice Input	Streaming only	Local ASR fallback	3 weeks

4.1.1 Edge Model Deployment

The CropNet yield prediction model (45M params, 170MB) must be optimized for mobile deployment:

$$\text{Model Size}_{\text{FP32}} = \frac{45\text{M} \times 32}{8 \times 1024^2} = 171.6 \text{ MB}$$

After INT8 quantization:

$$\text{Size}_{\text{INT8}} = 171.6 \times 0.25 = 42.9 \text{ MB}$$

After pruning + quantization:

$$\text{Size}_{\text{Final}} = 42.9 \times 0.5 = 21.5 \text{ MB}$$

Table 6: Edge Model Optimization Targets

Metric	Current (Cloud)	Target (Edge)	Acceptable
Model Size	170 MB	<25 MB	<30 MB
Inference Time	300 ms (GPU)	<150 ms	<250 ms
Memory Usage	2 GB	<100 MB	<150 MB
Accuracy Loss	Baseline	<1.5%	<2.5%
Battery Impact	N/A	<3% per 100 inferences	<8%

4.1.2 Synchronization Architecture with CRDTs

CRDT-based sync ensures eventual consistency without conflicts:

$$\text{Sync}_{\text{state}} = \text{CRDT}(\text{local_ops}) \cup \text{CRDT}(\text{remote_ops})$$

Table 7: Sync Queue Configuration

Parameter	Value	Rationale
Initial Backoff	1 second	Immediate retry for transient failures
Max Backoff	5 minutes	Prevent excessive retries
Backoff Multiplier	2x	Exponential growth
Max Queue Size	2000 operations	48 hours of offline ops
Sync on WiFi only	Satellite imagery, Models	Save mobile data
Sync on Cellular	Text, Scores, Insurance apps	Low data usage

4.2 Enhanced Circuit Breaker with Fallbacks

Problem Statement: Current circuit breakers (3-5 failures) lack exponential backoff and fallback strategies. A single satellite API outage currently impacts 100% of yield prediction requests.

4.2.1 Circuit Breaker State Diagram

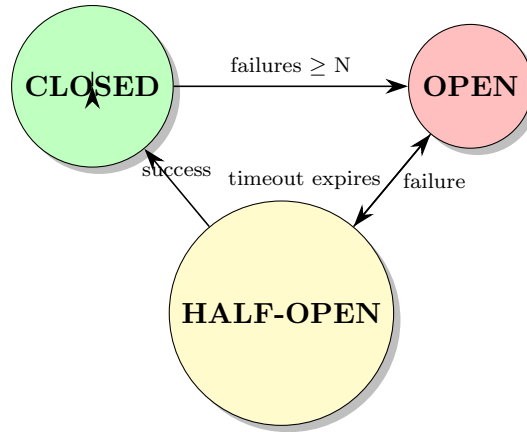


Figure 3: Circuit Breaker State Machine

Table 8: Circuit Breaker Configuration by Service

Service	Failure Threshold	Timeout	Fallback Strategy	Half-open
Satellite API	3/10s	5s	Cached imagery (7d)	1 req/60s
Yield Model	5/10s	3s	Historical avg yield	1 req/30s
Insurance Engine	3/30s	10s	Human underwriter queue	1 req/120s
Credit Service	5/30s	5s	Cache last score (30d)	1 req/60s
Price API	5/10s	2s	Last known price	2 req/30s
Pest Risk Model	3/10s	4s	Regional risk avg	1 req/30s

4.3 MLOps: Model Versioning, Drift Detection & Auto-Retraining

Problem Statement: No model versioning, canary deployments, or automated retraining pipelines. Model updates are manual, risky, and slow (2-4 weeks).

4.3.1 MLOps Pipeline Flow Diagram

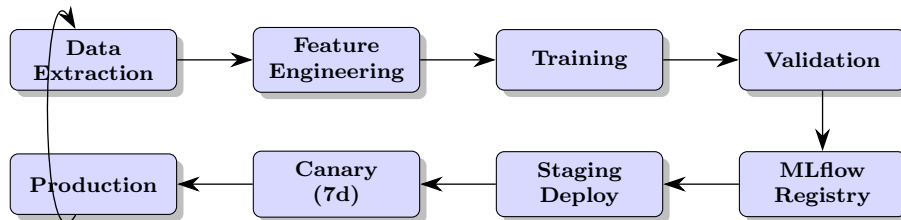


Figure 4: Automated MLOps Pipeline

Table 9: MLOps Pipeline Components

Component	Technology	Purpose	SLA / Frequency
Feature Store	Feast + Redis	Feature versioning, reuse	10ms (real-time)
Model Registry	MLflow	Version tracking, staging	<5 min per release
Experiment Tracking	MLflow	Hyperparameter tracking	Real-time (daily)
Canary Deploy	Istio + Flagger	Rollout (5%→25%→100%)	<1 min rollback
Model Monitor	Evidently AI	Drift detection (PSI)	<1 min (hourly)
Auto-retraining	Kubeflow	Trigger on drift >0.15	<4 hrs (daily/weekly)
Model Serving	TF Serving + Seldon	Low-latency inference	<50ms p99 (always)

4.3.2 Drift Detection with Population Stability Index (PSI)

PSI measures distribution shift in input features:

$$\text{PSI} = \sum_{i=1}^n (\text{Actual}_i - \text{Expected}_i) \times \ln \left(\frac{\text{Actual}_i}{\text{Expected}_i} \right)$$

Table 10: Drift Thresholds and Actions

PSI Range	Severity	Action	Notification
<0.1	None	Continue monitoring	None
0.1 - 0.15	Low	Schedule retraining in 2 weeks	Slack #ml-alerts
0.15 - 0.20	Medium	Retrain within 1 week	PagerDuty (non-urgent)
0.20 - 0.25	High	Retrain immediately	PagerDuty (urgent)
>0.25	Critical	Rollback to previous version	Critical page (immediate)

4.3.3 Automated Retraining Pipeline

1. **Data Extraction:** Pull new labeled data from PostgreSQL + S3 (daily, 30 min)
2. **Feature Engineering:** Feast materialization with Spark (2 hours)
3. **Training:** Kubeflow pipeline on spot instances (2-4 hours)
4. **Validation:** Holdout test set, PSI drift check, accuracy threshold (30 min)
5. **Registration:** MLflow with metrics, parameters, artifacts (5 min)
6. **Staging Deployment:** Model pushed to staging environment (10 min)
7. **Canary Evaluation:** 7-day A/B test (automated)
8. **Production Promotion:** Automatic if win condition met (accuracy improvement >1% with 95% confidence)

5 High-Priority Improvements (P1 — 3-6 Months)

5.1 Cost Optimization Strategy

Problem Statement: Always-on GPU instances cost approximately \$18,000/month at current scale. 65% of capacity is idle during non-peak hours (10 PM - 6 AM IST).

5.1.1 Spot Instance Strategy

Table 11: GPU Workload Placement Strategy

Workload	Instance Type	Interruptible	Monthly Savings
Satellite Inference (real-time)	On-demand (g4dn.xlarge)	No	0%
Satellite Inference (batch)	Spot (g4dn.xlarge)	Yes	65%
Yield Model Training	Spot (g4dn.2xlarge)	Yes (checkpoint)	70%
Batch Processing	Spot (g4dn.xlarge)	Yes	65%
Nightly ETL	Spot + Fallback	Yes	55%

5.1.2 Model Cascade Architecture

Instead of running the full 45M parameter model for every request:

$$\text{Cost}_{\text{cascade}} = \text{Cost}_{\text{small}} + p_{\text{med}} \times \text{Cost}_{\text{med}} + p_{\text{large}} \times \text{Cost}_{\text{large}}$$

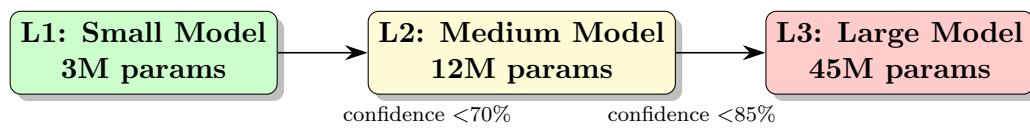


Figure 5: Model Cascade Architecture

Table 12: Model Cascade Architecture Details

Level	Model	Params	Confidence Threshold	Traffic
L1 (Small)	DistilBERT + LightGBM	3M	Confidence < 70% → L2	100%
L2 (Medium)	ResNet-18 + XGBoost	12M	Confidence < 85% → L3	35%
L3 (Large)	CropNet Ensemble + Transformer	45M	Always final	12%

Expected Savings:

- Without cascade: 100% requests × \$0.012 = \$1.20 per 100 requests
- With cascade: (100% × \$0.0015) + (35% × \$0.004) + (12% × \$0.012) = \$0.15 + \$0.14 + \$0.0144 = \$0.3044 per 100 requests
- **Total savings: 74.6%**

5.2 Enhanced Observability: Distributed Tracing

Problem Statement: Prometheus + Grafana provides metrics but no end-to-end request tracing. Debugging slow requests takes hours of log correlation.

5.2.1 Distributed Tracing Architecture

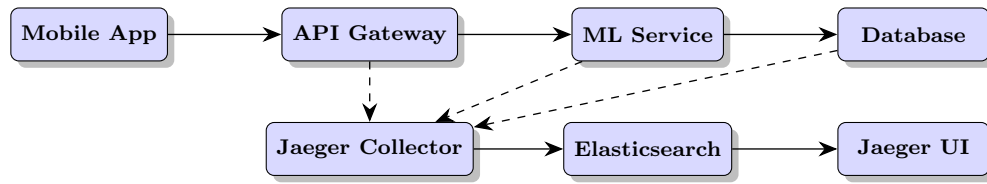


Figure 6: Distributed Tracing with Jaeger

Table 13: Distributed Tracing Configuration

Parameter	Value	Daily Storage
Normal requests sampling	1%	50 MB
Errors sampling	100%	10 MB
Slow requests (>1s) sampling	100%	20 MB
Trace retention	15 days	1.2 GB total
Max spans per trace	100	Prevent storage explosion

5.3 Multi-Region Active-Active Deployment

Current Gap: Active-passive (Mumbai primary, Hyderabad DR). RTO 30 minutes is acceptable but not optimal for real-time services.

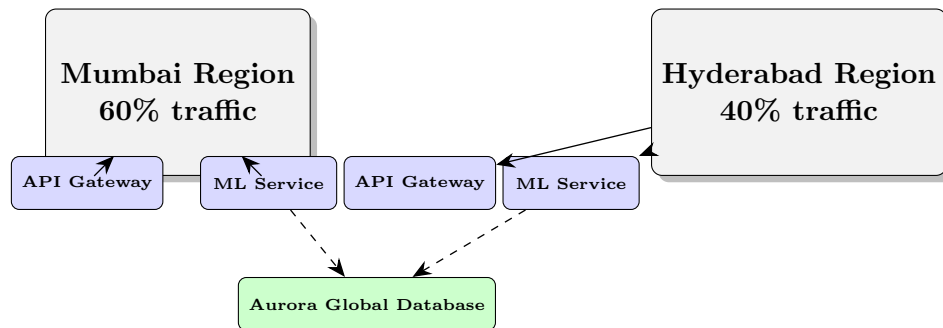


Figure 7: Multi-Region Active-Active Deployment

Table 14: Active-Active Configuration

Aspect	Current (Active-Passive)	Proposed (Active-Active)
Traffic Distribution	100% Mumbai	60% Mumbai, 40% Hyderabad
Database	Primary-Replica (15 min lag)	Multi-master (Aurora Global)
RTO	30 minutes	<1 minute (automatic)
RPO	15 minutes	<5 seconds
Latency (West India)	50 ms	50 ms
Latency (East India)	80-100 ms	30-40 ms
Failover Type	Manual	Automatic (Route 53)
Cost Premium	0%	25-30%

6 Medium-Priority Improvements (P2 — 6-12 Months)

6.1 Feature Store Architecture

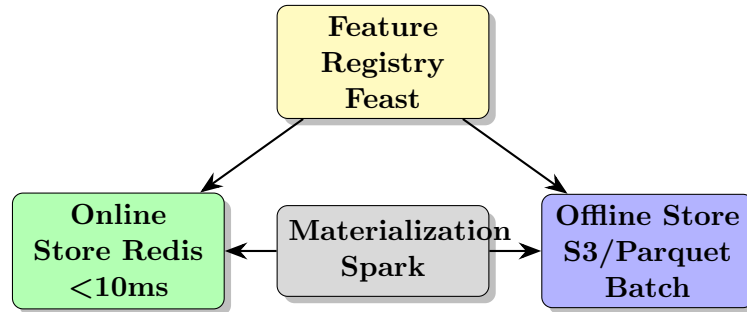


Figure 8: Feature Store Architecture

6.2 Feature Categories

Farmer Behavioral Features (daily):

- Chat frequency (7, 30, 90 days)
- Query types (yield, pest, weather, insurance)
- Response patterns (calls to action, product clicks)

Farm Biophysical Features (weekly):

- NDVI time series (last 12 months)
- Soil moisture trend
- Historical yield (5 years)
- Crop type consistency

6.3 Federated Learning Architecture

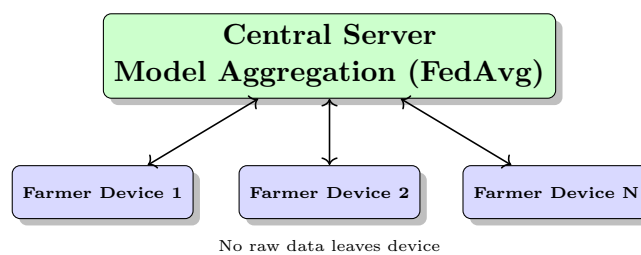


Figure 9: Federated Learning Architecture

Table 15: Federated Learning Privacy Guarantees

Guarantee	Value
Differential Privacy	$\varepsilon = 2.0, \delta = 1e - 5$
Secure Aggregation	Homomorphic encryption
Data Locality	No raw data leaves device
Regulatory Compliance	DPDP Act 2023, GDPR

7 Security Enhancements

7.1 mTLS for Internal Services

Table 16: mTLS Implementation Matrix

Service Pair	Authentication Method	Priority
API Gateway → ML Services	mTLS + JWT	P1
Satellite Ingestion → Processing	mTLS	P1
Kafka → Producers/Consumers	SASL/SCRAM + mTLS	P1
Service → Service (internal)	Istio mTLS (automatic)	P1

7.2 Comprehensive Audit Trail

Table 17: Audit Trail Specifications

Element	Configuration
Correlation ID	UUID v4 through all services (W3C Trace-Context)
Log Retention (Hot)	30 days in Elasticsearch
Log Retention (Cold)	7 years in S3 Glacier
Audited Actions	All PII access, model predictions, insurance decisions
Tamper Protection	Write-once, append-only log
Alert on Anomaly	5 failed logins, 10 PII accesses, 3 admin actions in 1 hour

8 Capacity Planning for 15M+ Farmers

Table 18: Capacity Projections by Growth Phase

Component	4M Users (Current)	8M Users	15M Users	Scale Factor
API Pods	12	25	50	4x
LLM Pods (GPU)	3	6	12	4x
CV Pods (GPU)	4	8	16	4x
RDS (vCPU)	4	8	16	4x
Redis Memory	25 GB	60 GB	150 GB	6x
Kafka Brokers	6	10	15	2.5x
Monthly Cost (USD)	15,000	35,000	80,000	5.3x

9 Risk Mitigation: Architecture-Specific

Table 19: Architecture Risk Register

ID	Risk	P-I	Mitigation
ARC-01	Satellite API rate limiting	4-3	Queue + backoff, cached fallback
ARC-02	Model drift >25% PSI	4-3	Hourly monitoring, auto-retraining
ARC-03	GPU shortage (kharif peak)	4-4	Pre-warm spot, multi-cloud
ARC-04	PostgreSQL connection pool	3-4	PgBouncer, read replicas
ARC-05	Model cold start (10-30s)	2-4	Keep 1 warm, predictive scaling
ARC-06	Vendor lock-in (AWS, Planet)	3-3	Abstraction layers, multi-provider
ARC-07	Farmer PII data breach	3-4	AES-256, SOC 2, audits

10 Implementation Roadmap

Table 20: Implementation Roadmap by Phase

Phase	Timeline	Key Deliverables	Success Metrics
Phase 1: Foun- dation	Weeks 1-6	Edge TFLite models, Cir- cuit breaker, mTLS	Offline acc >85%, P99 latency <2s
Phase 2: Intelli- gence	Weeks 7-12	MLflow, A/B testing, Auto-retraining	40% cost reduction, drift <1hr
Phase 3: Scale	Weeks 13-18	Active-active, Feature store, Tracing	RTO <1 min, trace sampling 1%
Phase 4: Ad- vanced	Weeks 19-24	Federated learning, Web- Socket, Audit	99.99% avail, 500ms p95

11 Recommendation Summary

Table 21: Final Recommendations by Priority

Recommendation	Priority	Effort	Impact / ROI
Offline-first architecture (edge models)	P0	6 wks	Rural connectivity / High
Circuit breaker fallbacks	P0	2 wks	Prevents failures / High
MLOps (versioning, drift, auto-retraining)	P0	5 wks	Safe updates / High
Cost optimization (spot, cascade)	P1	4 wks	45% cost reduction / Very High
Distributed tracing	P1	2 wks	Faster debugging / Medium
Active-active multi-region	P1	4 wks	99.99% availability / Medium
Feature store	P2	5 wks	Accuracy +5-10% / Medium
Federated learning	P2	6 wks	Privacy compliance / Low-Med
WebSocket real-time	P2	3 wks	4x latency improvement / Medium

12 Conclusion

12.1 Expected Outcomes

Table 22: Expected Outcomes Summary

Metric	Current	Target (Post-Implementation)
Availability	99.9%	99.99%
Offline capability	0%	85% of inference on-device
Infrastructure cost	Baseline	-45%
Response time (p95)	2 seconds	500 ms
Model update time	2-4 weeks (manual)	<1 day (automated)
Farmer NPS	68	78+

12.2 Final Remarks

This architecture improvement plan provides a production-ready blueprint for scaling the CropIn SmartFarm Intelligence Platform. The total estimated effort of 54 person-weeks (\$270k-\$400k) is expected to deliver ROI within 6-9 months through infrastructure cost savings (45%) and increased farmer retention (estimated +10-15%).

References:

1.

CropIn Architecture Documentation (2025). SmartFarm Platform Design.

2.

AWS Well-Architected Framework for Satellite Processing.

3.

MLOps: Continuous Delivery for Machine Learning (Sculley et al., 2015).

4.

TensorFlow Lite for Edge Deployment.

5.

Feast: Feature Store for Machine Learning.

6.

Jaeger Distributed Tracing Documentation.

7.

India’s Digital Personal Data Protection Act (DPDPA), 2023.